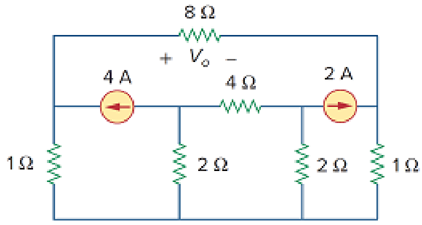
Chapter 3, Problem 24P

Problem

Use nodal analysis and *MATLAB* to find *Vo* in the circuit of Fig. 3.73.

Figure 3.73:



Step-by-step solution

Step 1 of 7

Consider the following circuit diagram:

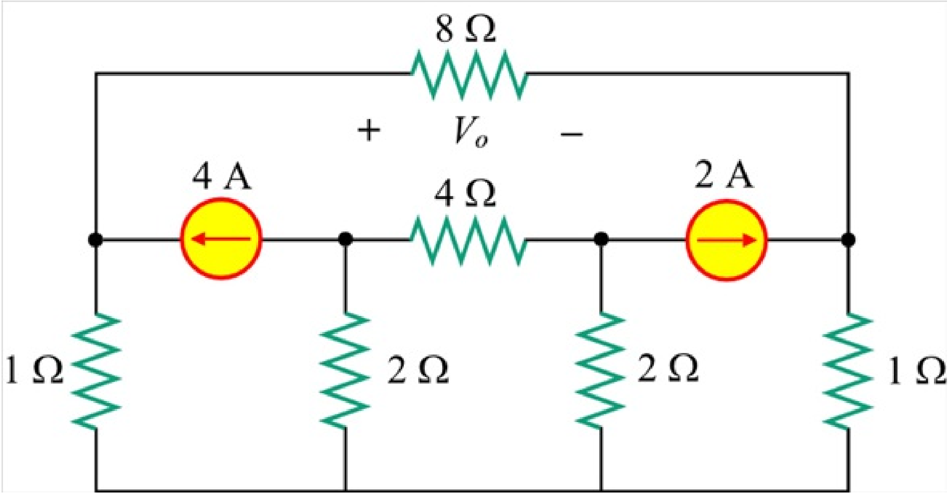


Figure 1

Step 2 of 7

Draw the circuit with node notations.

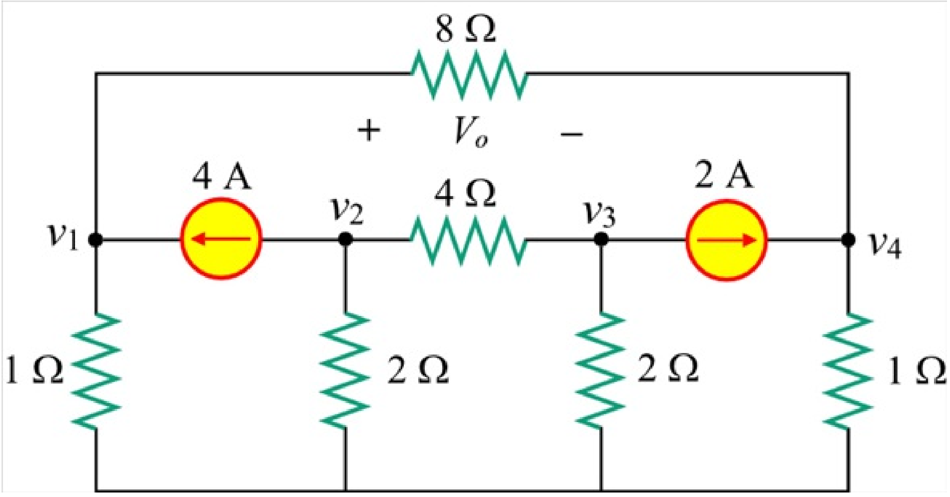


Figure 2

Apply Kirchhoff's current law at node 1.

$$\frac{v_1}{1}+\frac{v_1-v_4}{8}=4$$
$$\left(\frac{1}{1}+\frac{1}{8}\right)v_1-\frac{v_4}{8}=4$$
$$\left(\frac{8+1}{8}\right)v_1-\frac{v_4}{8}=4$$
$$(8+1)v_1-v_4=(4)(8)$$
$$9v_1-v_4=32 \quad \ldots\ldots (1)$$

Apply Kirchhoff's current law at node 2.

$$\frac{v_2}{2}+\frac{v_2-v_3}{4}=-4$$
$$\left(\frac{1}{2}+\frac{1}{4}\right)v_2-\frac{v_3}{4}=-4$$
$$\left(\frac{2+1}{4}\right)v_2-\frac{v_3}{4}=-4$$
$$3v_2-v_3=(-4)(4)$$
$$3v_2-v_3=-16 \quad \ldots\ldots (2)$$

Apply Kirchhoff's current law at node 3.

$$\frac{v_3}{2}+\frac{v_3-v_2}{4}=-2$$
$$\left(\frac{1}{2}+\frac{1}{4}\right)v_3-\frac{v_2}{4}=-2$$
$$\left(\frac{2+1}{4}\right)v_3-\frac{v_2}{4}=-2$$
$$v_2-3v_3=(2)(4)$$
$$v_2-3v_3=8 \quad \ldots\ldots (3)$$

Apply Kirchhoff's current law at node 4.

$$\frac{v_4}{1}+\frac{v_4-v_1}{8}=2$$
$$\left(\frac{1}{1}+\frac{1}{8}\right)v_4-\frac{v_1}{8}=2$$
$$\left(\frac{8+1}{8}\right)v_4-\frac{v_1}{8}=2$$
$$-v_1+9v_4=(2)(8)$$
$$v_1-9v_4=-16 \quad \ldots\ldots (4)$$

Write equations (1), (2), (3) and (4) in Matrix form.

$$\begin{bmatrix} 9 & 0 & 0 & -1 \\ 0 & 3 & -1 & 0 \\ 0 & 1 & -3 & 0 \\ 1 & 0 & 0 & -4 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}=\begin{bmatrix} 32 \\ -16 \\ 8 \\ -16 \end{bmatrix}$$

The following is the MATLAB code for calculating node voltages.

```
>>a=[9 0 0 -1;0 3 -1 0;0 1 -3 0;1 0 0 -9];>>b=[32;-16;8;-16];>>c=inv(a)*b
```

MATLAB output:

c =3.8000-7.0000-5.00002.2000

Step 7 of 7

The node voltages are,

$$v_1=3.8\text{ V}$$
$$v_2=-7\text{ V}$$
$$v_3=-5\text{ V}$$
$$v_4=2.2\text{ V}$$

Now calculate the value of voltage, V_o.

$$\begin{aligned} V_o &= v_1-v_4 \\ &= 3.8-2.2 \\ &= 1.6\text{ V} \end{aligned}$$

Therefore, the value of voltage, V_o is $\boxed{1.6\text{ V}}$.